

Assignment No 03

Name: Om Varma

Roll No: 28

PRN No: 1252010028

Question:

Static data member and static member function

Code:

```
#include<iostream>
using namespace std;

class BankAccount{
public:
    double* balance;

    static int totalAccounts;
    static double totalBalance;

    double validate(double b){
        if(b < 0) return 0;
        return b;
    }

    BankAccount(){
        balance = new double;
        *balance = 0;
    }

    ~BankAccount(){
        delete balance;
    }

    void input(){
        double b;
        cout << "Enter balance: ";
        cin >> b;

        *balance = validate(b);
        totalAccounts++;
        totalBalance += *balance;
    }

    void display(){
        cout << "Balance: " << *balance << endl;
    }

    static void showTotals(){
        cout << "Total accounts: " << totalAccounts << endl;
        cout << "Total balance: " << totalBalance << endl;
    }
};
```

```

    }

    static void showAscending(BankAccount a, BankAccount b){
        if(*a.balance < *b.balance)
            cout << *a.balance << " " << *b.balance << endl;
        else
            cout << *b.balance << " " << *a.balance << endl;
    }
};

int BankAccount::totalAccounts = 0;
double BankAccount::totalBalance = 0;

int main(){
    int n;
    cout << "Enter number of accounts: ";
    cin >> n;

    BankAccount* arr = new BankAccount[n];

    for(int i = 0; i < n; i++){
        cout << "Account " << i+1 << endl;
        arr[i].input();
    }

    cout << endl;

    for(int i = 0; i < n; i++){
        arr[i].display();
    }

    cout << endl;
    BankAccount::showTotals();

    if(n >= 2){
        cout << "Ascending of first two: ";
        BankAccount::showAscending(arr[0], arr[1]);
    }

    delete[] arr;
}

```

Output:

```

main.cpp powershell Assignment 3
(base) PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 3> .\main.exe
Enter number of accounts: 2
Account 1
Enter balance: 1002
Account 2
Enter balance: 2003

Balance: 1002
Balance: 2003

Total accounts: 2
Total balance: 3005
Ascending of first two: 1002 2003
(base) PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 3>

```